PREDICTING SONG POPULARITY FROM SPOTIFY AUDIO FEATURES: A TEMPORAL ANALYSIS OF CHANGING POPULARITY

Albert Skalinski
DKIT Dissertation

Contents

Abstract

This dissertation investigates to what extent Spotify audio features can be used to predict song popularity and whether the meaning of these features changes over time. The main research problem of this work is based on the question of what musical features drive song popularity. Two research questions were developed from that problem: how the structural determinants of song popularity evolved over time, and how accurately popularity can be predicted with the use of Spotify audio features.

The research uses a publicly available Spotify Music Dataset obtained from Kaggle. The original data was stored in two .csv files, which were joined and cleaned before the analysis. Missing values, duplicate rows, duplicate track IDs and problematic release dates were handled. Additional variables, such as releaseYear, releaseMonth, timePeriod and popularityGroup, were also created. After the cleaning process, the final dataset contained 4,368 songs.

The project followed a full Data Analytics pipeline. Firstly, Exploratory Data Analysis was conducted to investigate distributions, correlations and relationships between popularity and individual audio features. Then, a cultural and temporal analysis was added, where genre and time period were treated as contextual proxies. The modelling stage started with Linear Regression as a baseline model and XGBoost Regression as a more advanced non-linear model. The Linear Regression model achieved $R^2 = 0.065$, whereas XGBoost improved the result to $R^2 = 0.121$. Hyperparameter optimisation did not produce a meaningful improvement.

A contextual XGBoost model was then created by adding genre and releaseYear to the audio features. This model achieved $R^2 = 0.342$, showing a clear improvement over the audio-only models. Temporal models were also created for pre-2000, 2000s, 2010s and 2020s. Their permutation importance results showed that the importance of audio features changed between periods. Acousticness and valence were more important in older periods, whereas loudness became the most important feature in the 2010s and 2020s.

Overall, the results show that Spotify audio features have some predictive value, but by themselves they are not enough to explain song popularity accurately. The results also suggest that broader context, especially genre and release year, improves predictive performance. Finally, the temporal analysis shows that the relationship between audio characteristics and popularity is not stable and changes depending on the period.

Introduction

Background and Problem Definition

Music popularity is a phenomenon that is difficult to explain unambiguously. The success of a given song depends not only on its musical characteristics, but also on many external factors, such as artist popularity, promotion, being on popular playlists, social media trends, or the way streaming platforms work in general. It means, that popularity should not be treated as a result of the structure of the song alone. At the same time, modern streaming platforms, such as Spotify, allow analysing music with the use of quantitative data. It means, that using Data Analytics or Machine Learning methods is possible in many cases.

Spotify describes its songs by using measurable audio characteristics. The dataset that was analysed for the purpose of this paper contains variables such as danceability, energy, loudness, acousticness, instrumentalness, valence, speechiness, liveness, tempo, or the length of a given track. These characteristics represent various musical properties and can be used as predictors in popularity analysis. The target variable of this project is popularity, described as a quantitative value that refers to the popularity of a given song in the Spotify environment.

The main research problem of this work can be presented as the following question: what musical features drive song popularity? Although, the goal is not to assume that audio features alone explain the success of a song. As a matter of fact, one of the most important elements is to check how popularity can be explained with the use of audio features combined with additional context, i.e., the genre and the release year.

Another important factor of the problem is time. Popular music is not persistent: dominant genres, production techniques, or characteristics change. Because of that, dependencies between audio features and popularity might also differ between time periods. This project treats popularity as both a prediction and an interpretation problem, analysing not only accuracy of the models, but also temporal changes of certain features.

An important limitation is the fact that popularity is a variable connected strictly to Spotify and should be not automatically treated as a universal indicator of historical or cultural meaning of a song. The results of this project refer only to the analysis of the dataset taken directly from Spotify.

Research Problem and Relevance

The problem analysed in this work is important, mostly because the popularity of a song has an artistic, commercial and analytical nature. From the music industry point of view, knowledge about characteristics connected to popularity might be important for artists, producers, labels, or streaming platforms. It might support decisions of producing music, promoting, segmenting the audience, or analysing trends. Although, it does not mean that popularity can be simplified to a simple set of rules. Success of a song is also dependent on variables, that the analysed dataset does not contain, such as artist popularity, promotional budget, playlist positioning, or social media activity.

From the Data Analytics point of view, this problem is important, because it allows to implement a full analytical pipeline for cultural data. The project involves data cleaning, feature engineering, Exploratory Data Analysis (EDA), data visualisation, building regression models, judging them on their evaluation metrics, and the analysis of feature importance. Due to these factors, this paper is not limited to predicting the popularity value only but also connects the predictive approach with the result interpretation.

One thing is especially important – differencing between simply predicting popularity and understanding which features are useful for a model. The initial EDA showed that the majority of audio features do not show strong, linear relationships with popularity. Instrumentalness and acousticness were the only ones that stood out significantly – their relationship with the target variable was strong and negative. At the same time, there were strong correlations between two predictors themselves, especially energy and loudness. It suggested that popularity is a multidimensional phenomenon and needs a more complex model, rather than a simple correlation analysis.

The importance of the project is also increased by time and context. The cultural analysis showed that the genre structure of the dataset changes between time periods, and singular audio features do not stay persistent. The older periods are dominated by rock, whereas the newer ones are characterised by a larger genre variety. At the same time, one can notice that valence and track length decreased over time and instrumentalness, speechiness and danceability increased in more recent periods.

For the following reasons, the research problem is not only whether you can predict popularity or not, but also whether audio features stay stable over time. Such an approach allows to combine Machine Learning with the analysis of structural changes in contemporary music and builds a foundation for answering the research questions of the project.

Research Aim and Research Questions

The main goal of this research is to find out to what extent audio features available from Spotify can be used in explaining and predicting song popularity and whether the meaning of those features changes between different periods of time. The project then combines two approaches: predictive modelling and temporal interpretation. The first one concentrates on evaluating the accuracy of models predicting popularity, whereas the second one on the analysis of changes in the meaning of respective audio features.

The following research question has been formulated:

What musical features drive song popularity?

Based on that, two research questions have also been formulated:

RQ1: How have the structural determinants of song popularity evolved over time in contemporary music?

RQ2: How accurately can song popularity be predicted using Spotify audio features?

The first research question involves changes in time. It assumes, that features important for predicting popularity may not stay stable, because music genres, production techniques and the general structure of music changes over time. It was researched in the further parts of the project by building separate models for different periods and comparing their feature importance.

The second research question concentrates on the prediction accuracy. A linear Regression model was used as a basis, and an XGBoost model was used as a more advanced non-linear model. Then, an additional model that uses more variables (genre, releaseYear) was developed to check whether additional context improves the predictive performance.

This way both research questions are mutually inclusive: one contains information on how well popularity can be predicted, whereas the other whether the meaning of predictors changes over time.

Research Approach

To answer both research questions, this project uses a multi-step Data Analytics pipeline. The process started with preparing and cleaning the data. Two initial .csv files has been joined, and then missing values, duplicate rows, duplicate track IDs, datatypes, and release date correctness were all checked. Next, new time features were introduced, including releaseYear, releaseMonth, and timePeriod. Also, redundant columns were removed to simplify the structure of the dataset.

Subsequently, Exploratory Data Analysis was conducted, which contained the analysis of variations, correlations and relations between popularity and audio features. EDA was also used to identify the limitations of the dataset: it contained way more songs from 2010s and 2020s compared to the rest of the periods, as well as weak relationships between the majority of individual audio features with popularity.

After the interim submission, the pipeline has been expanded to include cultural and temporal analysis. In this part the following were analysed: changes in the genre variable structure, mean genre popularity, and audio characteristics in different time periods. Genre and time period were treated as contextual proxies, which allowed to include a broader musical context without stating that the dataset directly measures the culture.

The predictive modelling stage started with building a Linear Regression model as a baseline. Next, an XGBoost Regression model was built to see whether a non-linear model shows the relationships between popularity and audio features in a better way. Additionally, a hyperparameter optimisation was utilised with the use of RandomizedSearchCV. The next step was expanding XGBoost with genre and releaseYear variables, which allowed to compare the previous model that was only based on audio features with a model that included additional context.

The last stage was temporal modelling. The dataset was divided into four broader periods: pre-2000, 2000s, 2010s and 2020s. For each period, a separate XGBoost model was trained with the use of the same audio features. Next, the importance was calculated. That allowed to compare the meaning of respective predictors between time periods and directly address the first research question.

The entire pipeline combines three elements: data preparation, prediction, and interpretation, which allows to not only compare the accuracy of the models, but also investigate changes in time.

Structure of the Dissertation

This dissertation has been divided into a couple of main parts, which respectively correspond to stages of this research. In the next chapter, a literature review is presented. Its goal is to describe earlier research on predicting song popularity, the usage of audio features, temporal changes, and the role of broader genre and cultural context.

Subsequently, the chapter called Data and Methods describes the source of the data, structure of the dataset, process of data cleaning, feature engineering, EDA, cultural and temporal analysis and building the models. In this chapter, evaluation metrics, hyperparameter optimisation, and permutation importance are also described.

The next chapter called Results presents the most important EDA, cultural analysis and predictive modelling results. It includes the comparison of Linear Regression, XGBoost, the contextual model and finally the models built for respective time periods.

In the Discussion chapter the results are interpreted in the context of the research questions and previous literature. There is an emphasis on the limited predictive power of audio features themselves, the improved performance after adding genre and releaseYear variables, and changes in the meaning of certain audio features over time.

The last chapter, Conclusion and Future Work, summarises the most important results, answers both research questions and shows the biggest project limitations. It also prompts possible further research on the topic.

Literature Review

Music Popularity as a Data Analytics Problem

The popularity of music is a tough analytical problem, because there is no such thing as one characteristic that determines whether a song will be successful, or not. Popularity is based on factors such as characteristics of the song, but also on external factors, such as artist popularity, promotion, playlist presence, or streaming platform environments. From the Data Analytics perspective it means that it is a problem in which many factors can influence the target variable value.

The development of streaming platforms allowed to describe music in a form of quantitative data. Audio features present the characteristics of a song, such as energy, danceability, loudness, valence, acousticness, or instrumentalness in a form that is appropriate to use in statistical analysis and Machine Learning. Due to these factors, one can investigate not only relations between these features and popularity, but also how important they are in predictive modelling.

Interiano et al. (2018) show, that the presence of measurable characteristics does not mean, that musical success can be easily predicted. The authors stated that they were not able to exceed a prediction accuracy of 0.74. The result shows, that audio characteristics have predictive value, but are not the full explanation for popularity.

Saraigh (2023) came to a similar conclusion and states that both of his Machine Learning models yield a predictive power of 69%. It shows that Machine Learning might be efficiently used in music popularity prediction, but its results should be interpreted as a partial explanation of a more complex phenomenon. Such an approach forms an analytical foundation for the rest of this research.

Audio Features and Popularity Prediction

One of the main research directions on music popularity is checking if measurable audio features can be used for predicting the success of a song. In this approach variables such as loudness, energy, danceability, acousticness, instrumentalness, or valence are treated as predictors, where popularity or music chart presence are target variables.

Interiano et al. (2018) analysed over 500,000 songs from Great Britain released between 1985 and 2015, comparing the songs that appeared on charts with the ones that did not make it. The study showed that successful songs were different from the rest in terms of the musical characteristics, but the audio characteristics alone were not enough to make very accurate prediction. After adding information about previous artist popularity, the accuracy rose from 0.74 to 0.86, which shows the meaning of external factors.

Saragih (2023) analysed over 90,000 observations connected to Spotify users in Indonesia. The best regression model achieved R^2 = 0.6857 and RMSE = 12.33. Among the most important features were release date, loudness, instrumentalness, acousticness, duration, speechiness and energy.

Zangerle et al. (2019) also showed that prediction may be more accurate if combined with different types of information. Their model utilised both low-level and high-level audio features. They stated that “the release year information is the most important high-level feature” (p. 324), which furthermore emphasises on the temporal context in predicting popularity.

Overall, the results of these studies suggest that audio features have a real predictive value, although their accuracy rises after including additional context information.

Temporal Changes in Music

Another important element of the study of music popularity is changes in time. Features related to the success of a song may not be constant, because as the time passes, production techniques, dominant genres and audience preferences change. Because of that, analysing one global model might not be sufficient to fully understand the relationship between audio features and popularity.

Interiano et al. (2018) showed that the characteristics of popular songs changed over analysed time. In the study the noticed changes in features such as happiness, brightness, or sadness, which suggests that the characteristics are not stable over time.

The temporal context is also important on the genre level. Petitbon and Hitchcock (2022) analysed the changes in genre popularity between 1974 and 2018. One of the observed trend was the decline of rock popularity in the late 1980s through 2000s. Such inclines or declines in popularity might also affect the characteristics present in popular music.

The significance of time also appears in Zangerle at al.’s (2019) work, where release year was specified to be an important high-level feature. Overall, the results of these studies support the approach in which popularity prediction should be supplemented with temporal analysis, and feature importance should be compared between different time periods.

Genre and Cultural Context

The popularity of music is not dependent on characteristics of one song only, because music functions in a broader genre and cultural context. Musical genres might represent different styles of production and sound structure, and their popularity might change with the preferences of listeners. Because of that, genre might deliver an additional context for the popularity analysis, which audio features alone do not represent.

Petitbon and Hitchcock (2022) analysed the genre popularity, utilising data from *Billboard Hot 100* and *Village Voice Pazz & Jop*. The authors emphasised on a broader character of their approach and stated: “our analysis will focus on cultural, societal, and demographic factors that influence genre preference across a national culture” (p. 169). Their results show that the role of respective genres changes over time, as can be seen in the fall of popularity of rock and a rise of popularity of hip-hop.

At the same time, genre should not be treated as a direct and objective measure of culture. Petitbon and Hitchcock (2022) note that the genre classification is imperfect, because songs can belong to more than one category, and their classification is dependent on the method of the interpreter. In the context of this research, genre is then treated as contextual proxy, which can help in analysing musical changes, but does not measure the culture directly. Such an approach is important in the later interpretation of cultural and temporal analysis.

Limitations of Existing Approaches

So far the studies show that predicting music popularity whilst using audio features has some significant limitations. The most important one is the fact that there is no information on external factors, such as previous artist popularity, promotion, playlist presence, or social media activity.

Another limitation is the way popularity itself is defined. In different studies the target variable might mean the music chart presence, belonging to a certain popularity class, or a quantitative Spotify popularity score. It means that the results of different models not always can be directly compared.

The context of the data used is also important. Saragih (2023) analysed the data of Indonesian Spotify users, and because of that his results might reflect that specific market. On the other hand, other studies show Billboard or nation-wide musical charts. Consequently, predictive performance and feature importance might heavily rely on the method used, but also on the source of the data, the timeline, and the definition of musical success.

These limitations suggest that the model results should always be interpreted carefully and always in the context of the dataset on which they were built.

Research Gap

The literature review shows that previous studies on music popularity might be divided into a few different directions. Some of the work concentrates mostly on predictive modelling and checking whether the audio features can predict popularity and/or being on music charts. Other work concentrates more on changes that happen over time, such as evolution of musical genres, audience preferences, or characteristics of popular music. There are also approaches that combine audio features with additional high-level variables, such as genre, release year or artist popularity.

The main research gap does not arise from the lack of music popularity studies, but from the limited number of analyses that combine prediction, temporal comparison and the interpretation of feature importance. The evaluation of model performance alone does not show whether the characteristics stay up to date in different periods. On the other hand, the analysis of changing genres or musical trends does not answer the question of how good one can predict the popularity score.

This dissertation combines both approaches. Firstly, the predictive value of Spotify audio features using Linear Regression and XGBoost is evaluated. Then, the meaning of the same features across different periods of time is compared. Additionally, genre and releaseYear are inspected to see if the predictions are better compared to the model with audio-only features.

This way, the project is not limited to the question of whether the popularity can be predicted but also analyses whether the structure of the information used in such prediction changes over time. This approach combines both research questions.

Data and Methods

Data Source and Collection

The data used in this work comes from a publicly available Spotify Music Dataset published on the Kaggle platform. The dataset contains descriptive song information and a set of Spotify audio features, which can be used to quantitatively analyse the data. The data has been downloaded in two .csv files.

The data was neither collected directly from the Spotify users nor through web scraping or APIs. The project utilises an existing dataset shared by a third-party website, so the analysis is based only on the files included on the website. It also means that the work is limited to the scope and quality of the data from the source dataset.

Initial Dataset Description

Before starting the data cleaning, the data was stored in two separate .csv files. The first file contained 1,686 observations, whilst the second had 3,145 observations. Both of them contained the same 29 columns. The structure of the variables was the same for both files, but the order of the columns was different in the second file, so they were normalised before concatenation. After running pd.concat(), one set was created with 4,831 records.

The initial variables contained descriptive information, Spotify IDs, playlist information and audio features. Amongst the descriptive variables there was data such as track_name, track_artist, track_album_name, track_album_release_date and playlist_genre. The dataset also contained technical IDs, such as track_id, track_album_id and playlist_id.

The most important audio features were the following: energy, tempo, danceability, loudness, liveness, valence, speechiness, instrumentalness, acousticness, mode, key, time_signature and duration_ms. The target variable initially was saved as track_popularity.

The initial inspection showed the presence of missing values, duplicate rows, duplicate track IDs and problems with the format of release dates. Therefore, data cleaning was needed before proceeding to EDA.

Data Cleaning

The process of data cleaning started with checking for missing values. Initially in the dataset there were columns/rows that contained singular missing values. Because their number was substantially smaller than the entire dataset, records with missing values were removed. Then, exact duplicates were checked. 43 exact duplicates were detected, which were also removed.

The next step was the control of duplicate track IDs. Even after removing the exact duplicates, the dataset still contained 293 repeating track IDs. To make each song being represented only once, each subsequent appearance of a song was removed, i.e., only the first appearances stayed in the dataset. After that the dataset contained 4,493 observations.

Subsequently, datatypes of some variables were changed. time_signature, track_popularity, mode, key and duration_ms were converted to integers. Additionally, duration_ms was converted to minutes by creating a new duration_min variable, rounded to two decimal places.

There was an emphasis on the track_album_release_date variable, as it was key for future temporal analysis. The dates were converted to the datetime variable type. As a result, 125 records with incorrect/incomplete dates were detected, which were removed from further analysis.

At the end, columns that did not contribute to the analysis as a whole were also removed from the dataset. That included URLs, technical IDs, playlist_name, track_album_name and playlist_subgenre. The rest of the variables were tidied up and renamed to make it easier to reference in the future analysis, e.g., track_popularity to popularity, or playlist_genre to genre. After finishing the cleaning, the dataset has been saved as a new .csv file and used in future stages.

Feature Engineering

After completing the data cleaning stage, additional variables were created that were used in further analysis. Based on releaseDate, more specific variables were isolated: releaseYear and releaseMonth. Next, based on releaseYear, the timePeriod variable was defined and divided into pre-1980, 1980s, 1990s, 2000s, 2010s and 2020s.

During the EDA stage a popularityGroup variable was defined to make classification and comparison of audio features simpler and clearer. Songs with popularity lower than 40 were classified as Low popularity, values between 40 and 69 were Medium popularity, and values 70 and above were named as High popularity. This variable was used only for descriptive analyses purposes and was not utilised as a predictor in regression models.

At the temporal modelling stage another variable called modelPeriod was created, diving the data into Pre-2000, 2000s, 2010s and 2020s. Such broader division was implemented due to the fact that initial older time period sample sizes were not sufficient for accurate analyses.

In the modelling stage, the genre variable was converted using one-hot encoding, so it could be used with numerical audio features in XGBoost models.

Exploratory Data Analysis

After preparing the dataset, Exploratory Data Analysis was conducted to understand the data structure, distribution of variables and correlations between audio features and popularity. The analysis contained descriptive statistics, distribution plots, correlation analysis and comparison between previously created popularity groups.

Firstly, the distribution of popularity, the scope of releaseYear and the number of songs in each time period were analysed. It allowed to identify an important temporal imbalance – the majority of observations came from 2010s and 2020s, whereas the older periods were represented by much smaller sample size. The genre distribution was also inspected to identify which genres are most present in the dataset.

Subsequently, the correlation matrix was calculated for popularity, releaseYear and the main audio features. That analysis allowed to identify simple linear relationships and potential correlations between the predictor variables. Additionally, scatter plots for individual audio features and popularity were created. That was to show potential clear individual correlations.

At the end, mean values of audio features for low, medium and high popularity groups were compared. EDA primarily acted as a diagnostic and preparation tool before the modelling stage to show the most important patterns and limitations in the data.

Cultural and Temporal Analysis

After finishing the initial EDA, a cultural and temporal analysis was conducted. Its goal was to extend the project in terms of the genre and temporal context. The dataset did not contain any direct cultural variables, genre and timePeriod were treated as contextual proxies and not direct measurements of culture.

The first step was the analysis of genre composition over time. To make the results more readable, the ten most occurring genres were left as separate categories, whereas the rest was joined into the “other” group. The percentage of overall contribution to the given time period was then calculated. That allowed to compare different time periods, even though the sample sizes were not equal.

The next step was the analysis of the mean popularity based on genre and timePeriod. To limit the influence of very small groups, the genre-period combinations containing less than five songs were eliminated from the interpretation.

The last part contained the comparison of audio characteristics over time. Ten audio features were standardised, which allowed to analyse their relative changes on the same scale, despite the initial scale differences in some variables. Then mean values for each time period were calculated. At the end, the oldest and the newest period were compared.

Predictive Modelling

The predictive modelling stage has been designed to answer the second research question on the accuracy of popularity predictions. Ten continuous audio features were selected for main modelling: energy, danceability, loudness, liveness, valence, speechiness, instrumentalness, acousticness, tempo and durationMin. Key, mode and timeSignature variables were skipped in the main models.

The dataset was split using a train/test split (80/20) with the random_state parameter of 42 for reproducibility. The training set contained 3,494 observations, whereas the test set contained 874. The distribution of popularity in both parts was similar.

The first model was Linear Regression, used as a baseline. Audio features were re-scaled using StandardScaler before the training. Then, XGBoost Regression was used to see if a non-linear model is better suited to the relations between the predictors and the target variable.

For XGBoost a simple hyperparameter optimisation was also conducted using RandomizedSearchCV and a 3-fold cross-validation. Then, an additional contextual model was built in which releaseYear, and genre variables were added. Genre was previously transformed using one-hot encoding, which increased the number of predictors to 46. To make all models comparable, the same training and test observations were used throughout the modelling stage.

Model Evaluation and Feature Importance

To evaluate the predictive performance, three standard metrics for regression problems were used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2. MAE shows the mean absolute error of the prediction, whereas RMSE penalises larger mistakes. R^2 was used to evaluate the explainable variation of the popularity in the model. The set of these metrics allowed to compare Linear Regression, base XGBoost, optimised XGBoost, and the contextual model.

To answer the first research question, different temporal models were created. Due to the small sample size in older time periods, four groups were created: pre-2000, 2000s, 2010s, and 2020s. The same set of audio features and the same XGBoost configuration were used for each period to ensure smooth result comparison.

The meaning of predictors was evaluated based on permutation importance. This method measures the change of performance of a model after randomly shuffling the value of a single feature. If after such shuffling the model performs way worse, it means that the given variable was more useful for the prediction. The importance was calculated independently for each period using n_repeats = 10 and then the results were combined into one table and one heatmap.

Software and Ethical Considerations

The entire analysis was conducted in the Jupyter Notebook environment using the Python language. To prepare and process the data, mainly pandas and numpy libraries were used. The visualisations were created with the aid of matplotlib and seaborn. The modelling stage used scikit-learn to train/test split, StandardScaler, LinearRegression, evaluation metrics, RandomizedSearchCV, and permutation_importance. The non-linear model was built with the aid of the xgboost library and the XGBRegressor class.

From the ethical point of view, the dissertation did not involve any user contact. It also did not process any personal data. The dataset is publicly available and contains information on songs. It does not contain any information on individual Spotify users, their listening history, or anything that allows their identification.

The interpretation of the target variable (popularity) is also important. It is a measure associated with the Spotify streaming platform, so it should not be treated as a universal indicator of cultural value or historical success of a given song. The results of this work should be interpreted within the scope of the used dataset and the context of the Spotify platform.

Results

EDA Results

After finishing the data cleaning stage, the final dataset contained 4,368 songs. The mean value of popularity was ~53.1, with the median of 53. The distribution of popularity was bimodal, with clear peaks around 50 and 70. The most songs fell into the medium popularity category which contained 2,093 observations. The high popularity category contained 1,169 songs, whereas the low popularity group had 1,106 of them.

EDA also showed a strong temporal imbalance. Out of all observations, 2,799 songs came from 2020s. Only 47 came from pre-1980 and 67 from 1980s. It means, that the modern music is greatly overrepresented in the dataset.

The correlation analysis (figure 1) did not show any strong linear relationships between the majority of audio features and popularity. The significant negative correlations included instrumentalness and acousticness. At the same time, the strongest relationship between the predictors themselves appeared between energy and loudness, with the correlation of ~0.8.

The comparison of popularityGroup also showed that highly popular songs had lower instrumentalness and acousticness on average, where the differences between the majority of the other features were smaller.

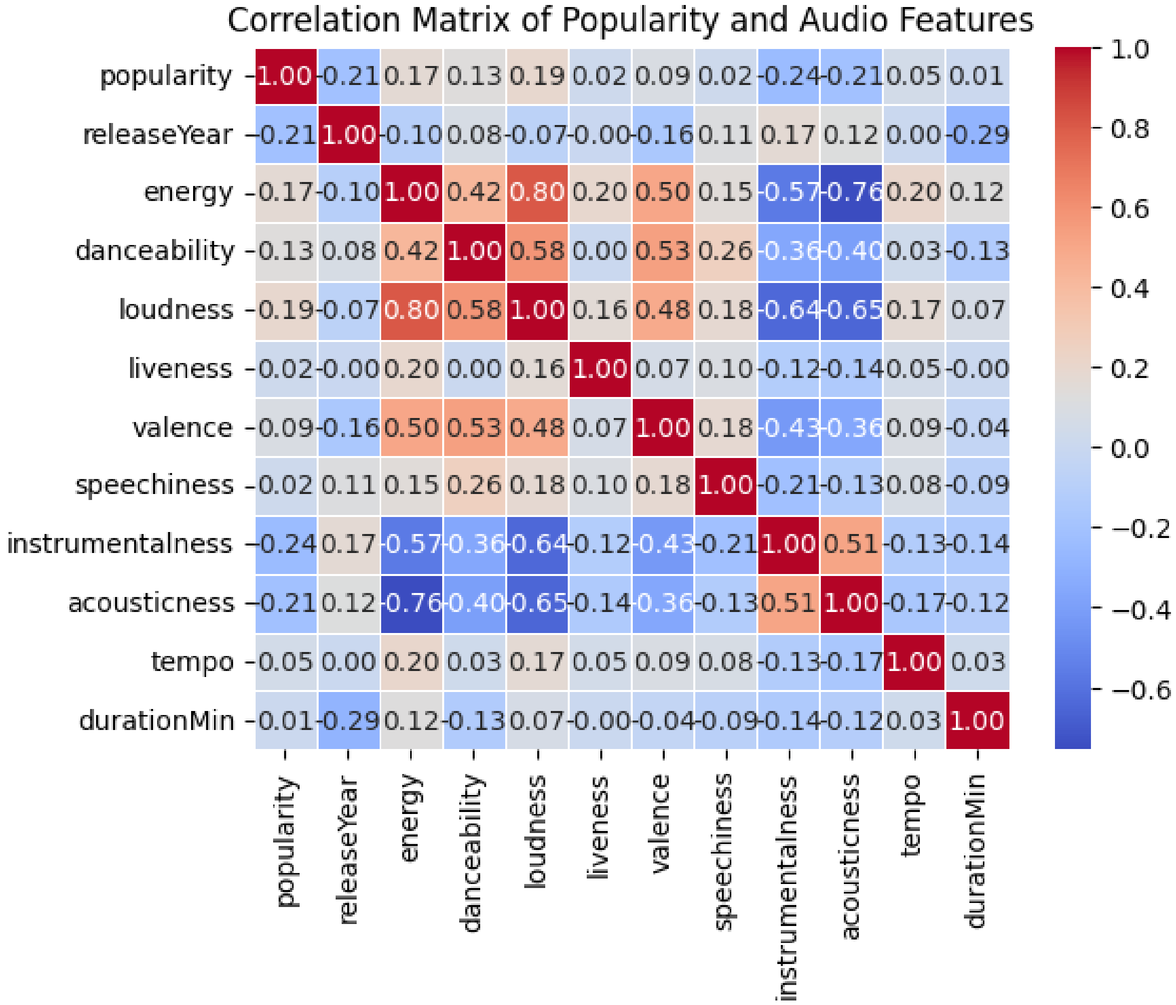

Figure 1 - Correlation Matrix of Popularity and Audio Features

Cultural and Temporal Results

The cultural and temporal analysis showed clear differences in the genre structures between time periods. Rock was dominant in the older periods. It represented ~66% of songs in pre-1980 and ~64% of songs from 1980s. In the next decades, the structure became more diverse. In 2010s, electronic and pop were the most represented. In 2020s, electronic, latin, lofi, ambient, and hip-hop were the most numerous (figure 2).

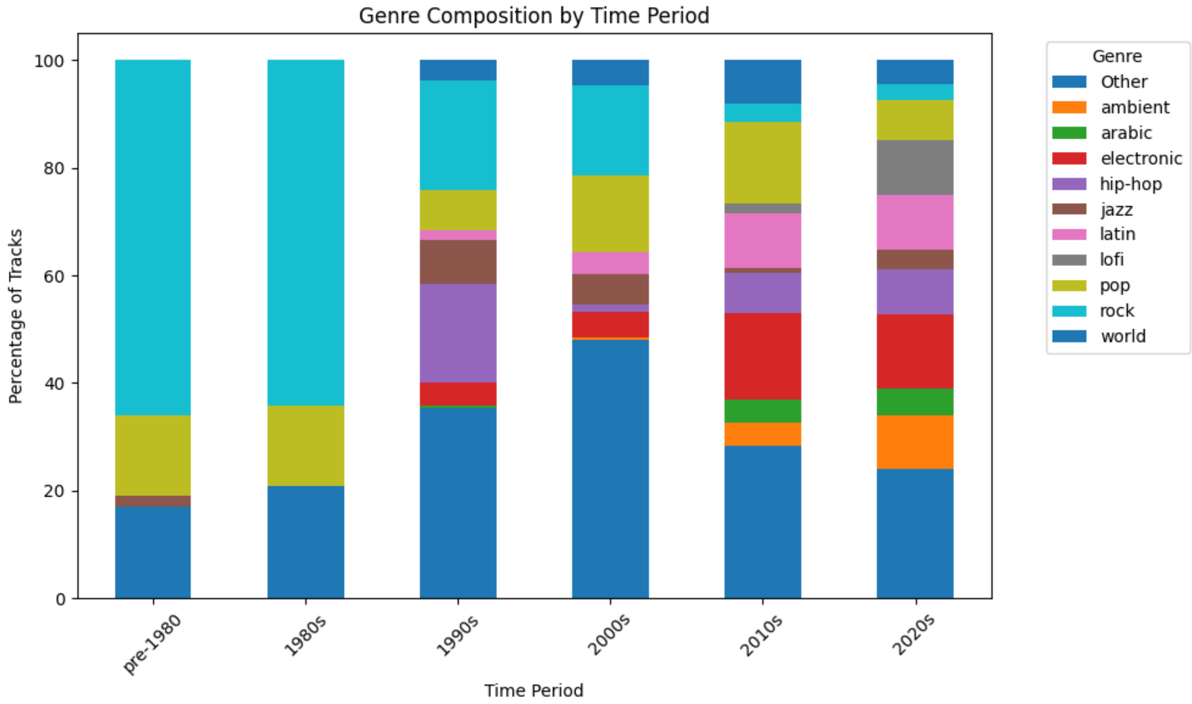

Figure 2 - Genre Composition by Time Period

The differences were also visible in the mean of popularity in respective time periods. For example, older pop and rock songs present in the dataset had higher Spotify popularity scores than their newer equivalents. This results have to be proceeded with caution though, because older time periods contain less observations and can be more prone to survivorship bias. In the genre-period analysis, combinations of less than 5 songs were excluded.

The temporal comparison of audio features also showed visible changes. Between pre-1980 and 2020s, valence and durationMin significantly dropped. On the other hand, a rise was observed in instrumentalness, speechiness and danceability. Energy changed little over time, but higher than normal values can be seen in 1980s (figure 3).

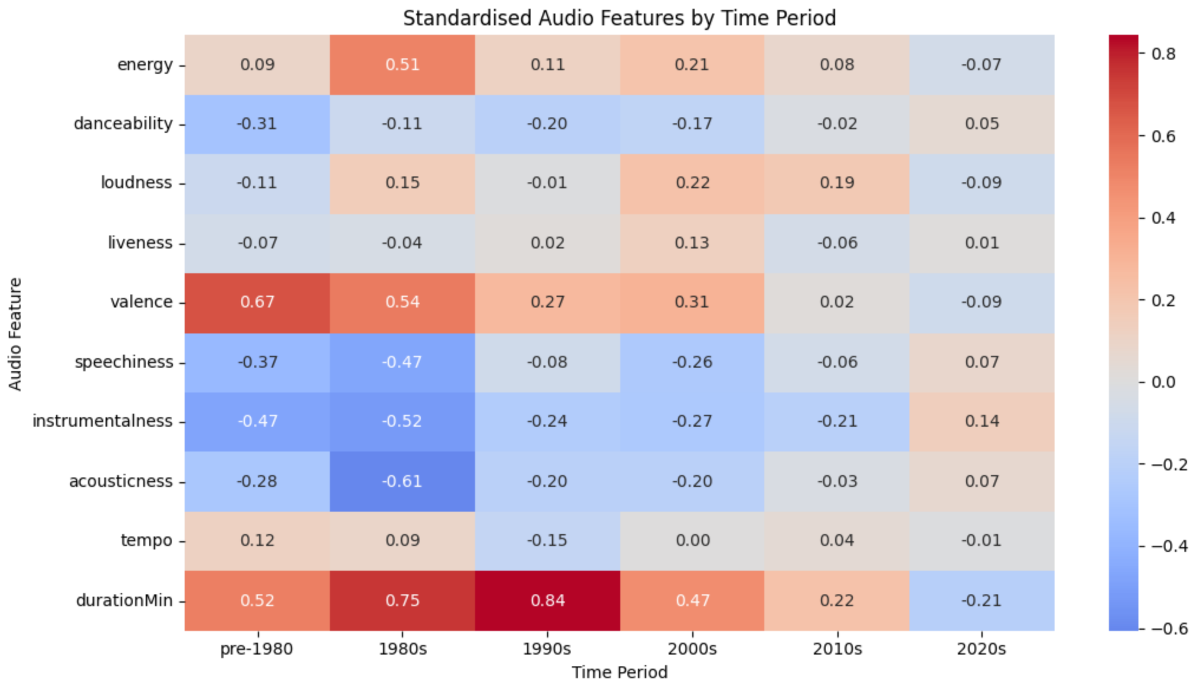

Figure 3 - Standardised Audio Features by Time Period

Baseline and XGBoost Performance

The first model was Linear Regression, which was the baseline model. It achieved the following results: MAE = 16.132, RMSE = 19.261 and R^2 = 0.065. It means that the linear model explained only a little part of the overall variation in popularity, and a mean prediction error was around 16 points of popularity.

Then, XGBoost Regression was implemented. The model achieved the following results: MAE = 15.201, RMSE = 18.673 and R^2 = 0.121. Compared to the baseline model, all three metrics improved. It means that the non-linear model showed the relations between the audio features and popularity.

After improving XGBoost with hyperparameter optimisation, it achieved MAE of 15.33, RMSE of 18.688 and R^2 of 0.12. The results were almost identical with the base XGBoost version, so the tuning did not improve on the predictive performance.

The results show that XGBoost had better predictions than the baseline linear model, but its predictive power stayed limited, due to containing audio features only (figure 4).

	Model	MAE	RMSE	R2
0	Linear Regression	16.132	19.261	0.065
1	XGBoost	15.201	18.673	0.121
2	Optimised XGBoost	15.330	18.688	0.120

Figure 4 - Model Comparison

Contextual Model Results

To check if additional context improves predictions, an XGBoost with the addition of releaseYear and genre was built. After implementing one-hot encoding the model contained 46 predictors in total. The same training and test observations as in the audio-only model were used.

The contextual model achieved the following results: MAE = 12.796, RMSE = 16.153 and R^2 = 0.342. Compared to audio-only XGBoost (which scored 0.12 in R^2), the increase is 0.223. MAE and RMSE decreased by ~2.53 points.

It was the largest performance improvement across all models. The results show that the information on genre and releaseYear were providing the model significant additional context. At the same time, the model still did not explain the majority of variation in popularity, so even after implementing the improvement, it stayed limited.

Temporal Model Performance

To see if the predictive performance changes between periods, different XGBoost models were built for pre-2000, 2000s, 2010s and 2020s. The results were significantly different between the groups.

The best results achieved the 2000s model with the following: MAE = 10.524, RMSE = 13.533 and R^2 = 0.383. The pre-2000 model achieved R^2 = 0.209, MAE = 11.042 and RMSE = 15.116. For more recent periods, the predictive performance was lower: the 2010s model achieved R^2 of 0.125 and the 2020s' R^2 was 0.149.

The results show that the dependency between audio features and popularity was not equally strong in all periods. The performance of the 2000s' model was especially better in comparison to 2010s and 2020s model.

One has to remember about the sample size, though. Pre-2000 contained 326 songs, 2000s 317, 2010s 926, and 2020s had them 2,799. Because of that, a direct comparison demands caution.

Feature Importance Across Time

The permutation importance analysis showed that the meaning of respective audio features changed between studied periods. For pre-2000, the highest importance had acousticness (1.775), and then valence (0.667) and liveness (0.398). In 2000s, acousticness was also dominant (3.919). Valance and energy were also relatively high: 1.652 and 1.397, respectively.

In more recent periods the structure of feature importance was different. For 2010s loudness was the most important (1.874), and then energy (0.484) and instrumentalness (0.424). In 2020s loudness was also dominant (1.544), which was followed by danceability (0.349), energy (0.294) and instrumentalness (0.235).

At the same time, the importance of acousticness and valence (which was high in the older periods) greatly decreased in 2020s. Negative values appeared for some features, e.g., danceability and speechiness in 2000s. It means the following: their random shuffling did not make the model perform worse and they were not useful predictors in the given model (figure 5).

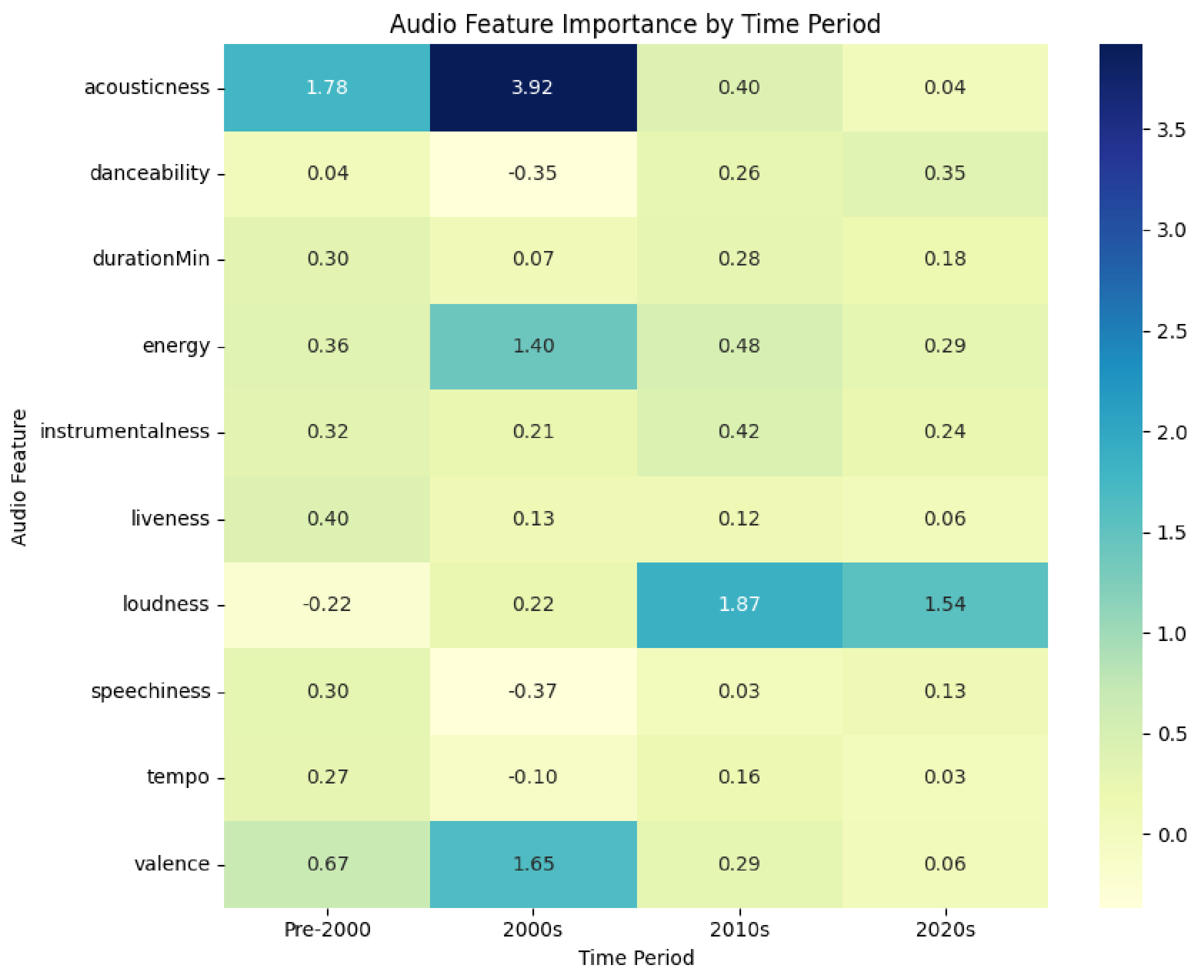

Figure 5 - Audio Feature Importance by Time Period

Discussion

Interpreting Predictive Performance

The predictive modelling results show that audio features alone have some kind of a predictive value, but their abilities are heavily limited. Linear Regression achieved R^2 = 0.065, whereas XGBoost improved on that score to ~0.121. It means that the non-linear model was better at capturing more complex relationships between audio features and popularity but still explains just a little part of the variation of the target variable.

It is also worth noting that the hyperparameter optimisation practically did not improve performance. Optimised XGBoost achieved R^2 = 0.12, so an almost identical score compared to the base model. It suggests that the main limitation were not poorly chosen parameters, but the amount of information within audio features themselves.

This result is consistent with the previous EDA, which did not show any strong simple relationships between the majority of single audio features and popularity. Popularity of a song seem to depend on a larger number of factors, not only on the musical characteristics. Audio features can support predictions, but by themselves are not enough to build a model with high predictive accuracy.

Importance of Temporal and Genre Context

The highest improvement in predictive performance occurred after expanding the model by adding genre and releaseYear variables. Contextual XGBoost achieved R^2 = 0.342, whilst the optimised audio-only model got only 0.12. It means that the additional cultural and temporal context delivered information that the audio features did not contain.

The result suggests that the popularity of a song is not only connected to the audio features, but also to the release period and a larger genre context. Genre might indirectly represent differences in style, production and music reception. ReleaseYear, on the other hand, might reflect the trend changes and the characteristics of Spotify popularity score.

Although, it does not mean that genre and releaseYear directly measure the cultural aspects. In the context of this work, they're treated as contextual proxies. releaseYear, especially, requires special attention, because older tracks are significantly less represented in the dataset and might involve songs, that have been popular for a long time.

In practice, the results of the contextual model show that the predictions based on audio features only skips an important context part associated with the genre and time.

How Popularity Determinants Changed Over Time

The permutation importance analysis showed that the meaning of particular audio features was not stable across time periods. In pre-2000 and 200s models acousticness played a bigger role. Valence and energy were also relatively high. In more recent years, the structure changed. For 2010s and 2020s loudness became the most important feature. Danceability also played a big role in 2020s.

This change suggests that the predictive relationship between the song characteristics and popularity depends on time. It does not mean that specific features cause popularity, but the model uses them to some extent based on a given temporal subset. In practice, it means that one global model might hide differences that are only visible after diving the model into different periods.

It is also worth noting that the predictive performance of the same temporal models was not the same. 2000s had the highest R^2, whereas scores for 2010s and 2020s were lower.

At the same time, comparing these periods require caution because of different sample sizes. Pre-2000 and 2000s contain less observations, so their feature importance might be less stable. Nevertheless, the general pattern supports the assumption that structural predictors of popularity change over time.

Cultural Interpretation

The results of cultural and temporal analysis show that changes in audio features occur collaterally with the changes in the genre structure of the dataset. Older periods were strongly dominated by rock, whereas there was more diversity in 2010s and 2020s (electronic, latin, hip-hop, ambient, lofi).

At the same time, the average characteristics of songs also changed. In more recent years valence and durationMin decreased. On the other hand, danceability, speechiness and instrumentalness increased.

One should not interpret these results as direct measurements of cultural changes, though. The dataset does not contain variables describing social values or audience demographics. Genre and time period are only contextual proxies here. The results might be treated as an evidence of changes in the structure of music represented by the dataset and not as a full explanation of cultural changes in the society.

Comparison with Previous Literature

The results of this project largely align with the previous studies presented in the Literature Review section. A limited accuracy of audio-only models corresponds to Interiano et al.'s (2018) results. In their work audio features only allowed for a prediction to some extent but did not give a high predictive performance.

Similarities can also be seen in relation to Saragih (2023). But direct comparisons his values with the results of this study are suitable, because both studies use different datasets, models and the ways to define popularity.

The result of the contextual model is especially important. In Zangerle et al.'s (2019) work adding releaseYear alongside genre also brought a significant improvement of performance compared to an audio-only model.

Temporal results are aligned with the general direction shown by Petitbon and Hitchcock (2022). It supports the interpretation that both the genre composition and the meaning of defined audio characteristics might change with time.

Limitations

The most important limitation is the structure of the dataset. Great majority of the songs comes from 2010s and 2020s, and older periods are represented by a smaller number of observations. It means that the results of temporal models for pre-2000 and 2000s might be less stable and more prone to be affected by individual songs.

The second limitation is the possibility of survivorship bias. Older songs that are currently available on Spotify playlists might the ones that have been popular for many years. Because of that the popularity of older songs should not be interpreted as an evidence that an average song released in that period of time was more popular.

Another problem is the lack of important external factors. The dataset does not contain and information on previous artist popularity, playlist placement, social media, marketing budget or actual number of streams. It might partially explain a limited performance of audio-only models.

Popularity is also a measurement specific for Spotify, so it cannot be treated as a universal measure of musical success. On top of that, genre in this case plays a role of a contextual proxy and is not a direct measure of culture.

Finally, all results are observational. Feature importance and model coefficients show predictive associations, not causal relations. Because of that one cannot state that a certain audio characteristic directly causes a decrease or an increase in popularity.

Answering the Research Question

The results of this study allow to answer both research questions. In terms of RQ1, structural determinants of popularity change over time. Permutation importance shows that in older periods acousticness and valence were more important, whereas more recently it was loudness and danceability. The predictive structure change with the time period.

In terms of RQ2, Spotify audio features allow predictions of popularity, but to some limited extent. The best audio-only model, XGBoost, achieved R^2 of ~0.12, which shows a weak predictive accuracy. After adding genre and releaseYear variables it increased to 0.342, showing that additional context improves the performance.

Final results support both assumptions of this dissertation: audio features have partial predictive value and their meaning changes over time.

Conclusion

The goal of this research was to investigate to what extent Spotify audio features might be used in predicting the popularity of songs and whether the meaning of these features changes over time. To accomplish this, a full Data Analytics pipeline was implemented, including data cleaning, feature engineering, EDA, cultural and temporal analysis and predictive modelling with the use of Linear Regression and XGBoost.

The results show that audio features alone do have limited predictive value. Linear Regression achieved R^2 = 0.065 and XGBoost improved that score to ~0.121. Hyperparameter optimisation did not improve the metrics. Much better results were achieved after expanding the model with genre and releaseYear variables, where R^2 increased to 0.342. It shows that contextual information gives the model significant value on top of the audio characteristics of the song.

The temporal analysis showed that the feature importance does not stay stable. Acousticness and valence were more prevalent in older periods, whereas in 2010s and 2020s loudness was more important.

Finally, the popularity of music cannot be fully explained just by its measurable audio characteristics. Although, the results show that these features have analytical value, and their predictive importance changes based on the time period and a broader analytical context.

Contribution

The main contribution of this dissertation is the combination of predictive modelling with the analysis of temporal and contextual changes into one coherent pipeline. The project is not limited to the evaluation of model performance only but uses feature importance for checking whether the meaning of individual audio features changes over time.

An additional value is extending the analysis of genre and releaseYear, which were treated as contextual proxies. It allowed showing that a model that includes a broader context achieves better results in terms of predictive performance compared to audio-only models.

As a result, this dissertation combines prediction, interpretability and temporal analysis, delivering a larger image of dependencies between the characteristics of music and its popularity.

Future Work

Future studies could, most importantly, use more balanced temporal dataset, which contains more songs from older periods. It would allow to limit the influence of uneven sample sizes and the survivorship bias, which make the feature importance between decades comparison harder.

Another extension would be adding external factors, which this dataset does not contain. It could be artist popularity, playlist placement, number of streams, social media activity, or information on promotions. Such variables could help in checking what part of popularity is connected to audio characteristics, and what part is connected to a broader social context.

In the future more Machine Learning models could also be implemented that utilise more detailed temporal division, but only if a larger number of observations for older periods is available. It would allow verification of whether the observed changes in predictive importance are the same in different models and datasets.

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The above references have been generated with the aid of this tool: https://www.mybib.com/tools/harvard-referencing-generator

Dataset used: https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset/data

The code used in this submission was generated with the aid of AI. All the links to conversations with ChatGPT are present in the artefact provided.